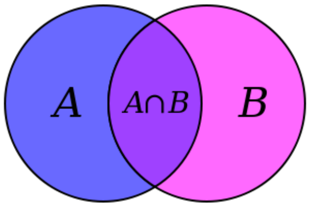


Set Theory:

SYMBOL	MEANING	EXAMPLE
$\{\}$	is a set	$S = \{4,5\}$
$\in$	is an element of	$s \in S$
$\notin$	is not an element of	$s \notin T$
$\subseteq$	is a subset of	$S \subseteq T$
$\subset$	is a proper subset of	$S \subset T$
$\cup$	union	$S \cup T$
$\cap$	intersection	$S \cap T$
$\varnothing$	the empty set	$\{2,3,4\} \cap \{5,6,7\} = \varnothing$

Topics of Study:

Set Notation: Roster notation, set-builder notation, interval Notation

Set Notation: Operations with sets

Union, Intersection, and Complement

Videos to introduce set theory including operations of sets:

Tutorials

1. Video for interval notation/set builder notation/roster notation
2. Video for interval notation
3. Introduction to sets: Intersection and Union
4. Video for all operations of sets

Extra practice/interactive links:

1. Types of sets and set notation practice
2. Union and Intersection of sets
3. Cumulative Review
4. Union and Intersection Extra Practice with Answer Key

Introduction to Set Theory: Set Notation

Set Theory: A set is a collection of unique elements. Elements in a set do not "repeat".

Methods of Describing Sets: Sets may be described in many ways: by roster, by set-builder notation, and by interval notation

Method 1: Roster Notation:

A **roster** is a list of the elements in a set, separated by commas and surrounded by French curly braces.

$\{2, 3, 4, 5, 6\}$	is a roster for the set of integers from 2 to 6, **inclusive** (including all numbers)
$\{1, 2, 3, 4, \ldots\}$	is a roster for the set of positive integers. The three dots indicate that the numbers continue in the same pattern indefinitely. The set continues towards positive infinity

Method 2: Set-builder notation:

Set-builder notation is a mathematical shorthand for precisely stating all numbers of a specific set that possess a specific property. (using inequalities)

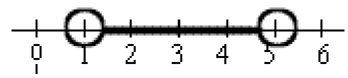
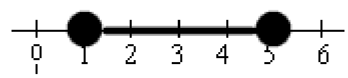
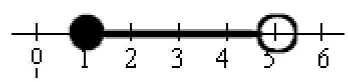
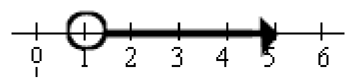
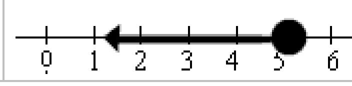
$\mathbb{R}$= real numbers; $\mathbb{Z}$ = integer numbers; $\mathbb{N}$= natural numbers.

$\{x \in \mathbb{Z} \mid 2 \le x \le 6\}$	is set-builder notation for the set of integers from 2 to 6, inclusive. $\in$= "is an element of" Z = the set of integers | = the words "such that" The statement is read, "all *x* that are elements of the set of integers, such that, *x* is between 2 and 6 inclusive."
$\{x \in \mathbb{Z} \mid x > 0\}$	The statement is read, "all *x* that are elements of the set of integers, such that, the *x* values are greater than 0, positive." (The positive integers can also be indicated as the set Z^+.)
It is also possible to use a colon (:), instead of the | , to represent the words "such that". $\{x \in \mathbb{Z} \mid 2 \le x \le 6\}$ is the same as $\{x \in \mathbb{Z} : 2 \le x \le 6\}$	

Method 3: By interval notation: An **interval** is a connected subset of numbers. **Interval notation** is an alternative to expressing your answer as an inequality. Unless specified otherwise, we will be working with real numbers.

When using interval notation, the symbol:
(means "not included" or "open".
[means "included" or "closed".

$2 \leq x < 6$	as an inequality.
$[2, 6)$	in interval notation.

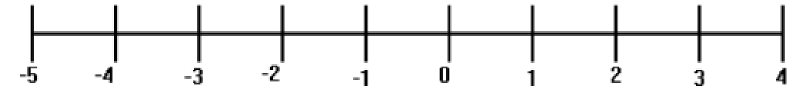
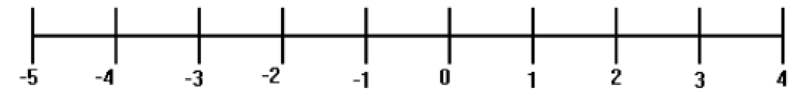
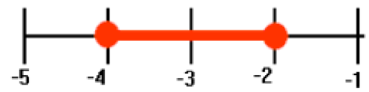
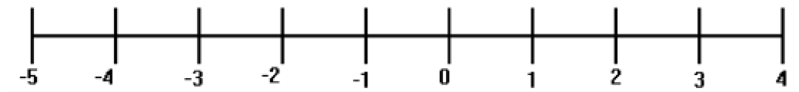
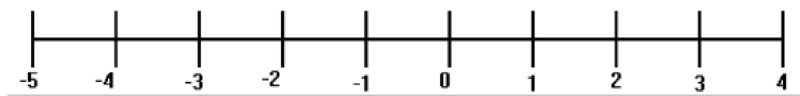
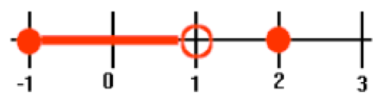
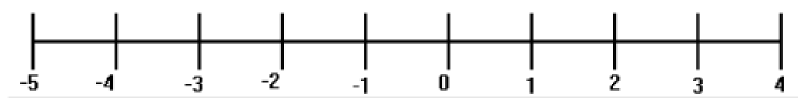
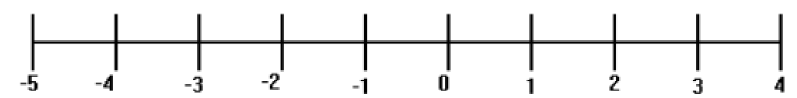
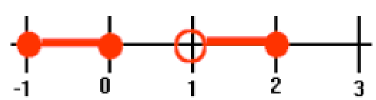
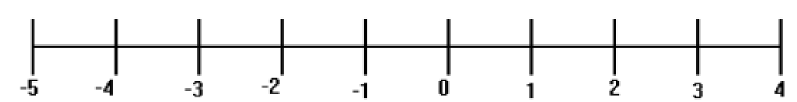
Interval Notation: (description)	(diagram)
Open Interval: (a, b) is interpreted as $a < x < b$ where the endpoints are NOT included. (While this notation resembles an ordered pair, in this context it refers to the interval upon which you are working.)	(1, 5) 
Closed Interval: $[a, b]$ is interpreted as $a \leq x \leq b$ where the endpoints are included.	[1, 5] 
Half-Open Interval: $(a, b]$ is interpreted as $a < x \leq b$ where a is not included, but b is included.	(1, 5] 
Half-Open Interval: $[a, b)$ is interpreted as $a \leq x < b$ where a is included, but b is not included.	[1, 5) 
Non-ending Interval: (a, ∞) is interpreted as $x > a$ where a is not included and infinity is always expressed as being "open" (not included).	$(1, \infty)$ 
Non-ending Interval: $(-\infty, b]$ is interpreted as $x \leq b$ where b is included and again, infinity is always expressed as being "open" (not included).	$(-\infty, 5]$ 

As an inequality:	$x < 0$ or $2 < x \leq 10$
In interval notation:	$(-\infty, 0) \cup (2, 10]$

<u>In Summary:</u>

The following statements and symbols ALL represent the same interval:	
WORDS:	**SYMBOLS:**
"all numbers between positive one and positive five, including the one and the five."	$1 \leq x \leq 5$
"x is less than or equal to 5 and greater than or equal to 1"	$\{ x \in \mathbb{R} \mid 1 \leq x \leq 5\}$
"x is between 1 and 5, inclusive"	$[1,5]$

Practice: Putting it all together:

	Set-builder Notation	Interval Notation	Graph
1.	$\{x \in \mathbb{R} \mid x < 3 or x \geq 5\}$		
2.		$(-4,0) \cup (1,\infty)$	
3.			
4.	$\{x \in \mathbb{R} \mid -5 \leq x < 4\}$		
5.		$(-\infty, 2) \bigcup (2, \infty)$	
6.			
7.	$\{x \in \mathbb{R} \mid x < -3 \text{ or } -2 \leq x \leq 3\}$		
8.		$(-\infty, -4] \cup (-2, \infty)$	
9.			
10.	$\{x \in \mathbb{R} \mid x \leq -2 \text{ or } -1 < x < 4\}$		

Operations of Sets

Remember: Sets are simply collections of items. A set may contain your favorite even numbers, the days of the week, or the names of your brothers and sisters. The items contained within a set are called **elements,** and elements in a set do not "repeat".

The elements of a set are often listed by **roster notation.** A **roster** is a list of the elements in a set, separated by commas and surrounded by French curly braces.

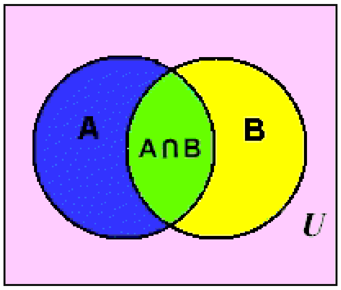
The **empty set is denoted with the symbol: $\varnothing = \{\}$

Examples:

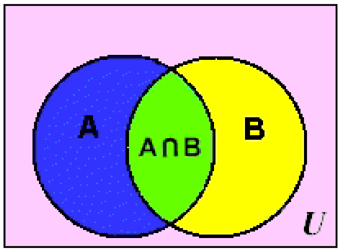
Let set A be the numbers 3, 6, 9. $A = \{3, 6, 9\}$ (in roster notation) $3 \in A$ The symbol, $\in$, is read "is an element of". $5 \notin A$ (5 is not in set A)	Let B = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} A = {3, 6, 9} Set A is a **subset** of set B, since every element in set A is also an element of set B. The notation is: $A \subset B$

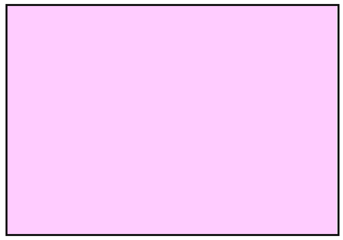
Operations:

1. **Intersection of sets:**
The Venn diagram below shows two sets ***A*** and ***B*** that overlap. The universal set is ***U***. Values that belong to both set ***A*** and set ***B*** are located in the center region labeled $A \cap B$ where the circles overlap. This region is called the "**intersection**" of the two sets.
(Intersection, is only where the two sets intersect, or overlap.)



2. **Union of Sets:** The notation $A \cup B$ represents the entire region covered by both sets ***A*** and ***B*** (and the section where they overlap). This region is called the "**union**" of the two sets.
(Union, like marriage, brings all of both sets together.)





3. **<u>Complement of Sets:</u>** If we cut out sets ***A*** and ***B*** from the picture above, the remaining region in ***U***, the universal set, is labeled , $(A \bigcup B)^c$ and is called the **complement** of the union of sets ***A*** and ***B***.
A complement of a set is all of the elements (in the universe) that are NOT in the set.

NOTE*: The **complement** of a set can be represented with several differing notations.
The complement of set A can be written as

$$A^c \text{ or } A' \text{ or } \overline{A} \text{ or } \sim A$$

Example:

Let U (the universal set) = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} (a subset of the positive integers)

$A = \{2, 4, 6, 8\}$

$B = \{1, 2, 3, 4, 5\}$

$A \bigcup B = \{1, 2, 3, 4, 5, 6, 8\}$ **Union** - ALL elements in BOTH sets

$A \bigcap B = \{2, 4\}$ **Intersection** - elements where sets overlap

$A^c = \{1, 3, 5, 7, 9, 10\}$ **Complement** - elements NOT in the set

$B^c = \{6, 7, 8, 9, 10\}$

Extra Practice of Complements:

Suppose that set $B = \{1, 2, 3, 4, 5, 6\}$ represents the "given universe" for a certain situation. Suppose, further, that set $A = \{2, 4, 6\}$. Then, we know that A is a subset of B. We express that relationship with symbols as $A \subset B$. The **complement** of set A here is $\{1, 3, 5\}$ because those are the elements in set B that are **not** in set A. In other words, the complement of a subset consists of all elements in the "given universe" that are not in the subset.

***Exercise* #2:** Consider the set E defined as $E = \{2, 3, 5, 7, 11, 13\}$ and set F defined as $F = \{2, 5, 7\}$. Determine the complement of set F within set E. (We denote the complement of set F using many different symbols: $\sim F$, F', or $\overline{F}$.)

***Exercise* #3:** Consider the set of integers as the "given universe." Determine the complement of the set of whole numbers. Express the complement in roster form.

***Exercise* #4:** What is the complement of the set that represents the "given universe" in any situation?

***Exercise* #5:** Given the set of numbers on a standard six sided die, which of the following represents the complement of the set comprised of all multiples of 3?

(1) $\{1, 2, 3\}$ (3) $\{1, 2, 3, 4, 5\}$

(2) $\{2, 4, 6\}$ (4) $\{1, 2, 4, 5\}$

____

Extra Practice with Union and Intersection:

Operations on Sets: Union and Intersection
Algebra 1

We don't "add" two sets together. Nor do we subtract, multiply, or divide them. However, there are two operations on sets that are commonly used. We call the operations **union** and **intersection.** Just as when we use basic operations on numbers we get numbers for answers, when we operate on sets we get sets for answers. Thus, the union of two sets is a set, and the intersection of two sets is a set. The definitions of union and intersection are given below in set-builder notation.

THE UNION AND INTERSECTION OF TWO SETS

For two sets, A and B, their union, $\cup$, and their intersection, $\cap$, are given by:

(1) **Union**: $A \cup B = \{x : x \in A \text{ or } x \in B\}$ (2) **Intersection**: $A \cap B = \{x : x \in A \text{ and } x \in B\}$

The union of two sets can be thought of as the "joining" of the sets (any element in either set must appear in their union). The intersection of two sets consists of whatever elements the two sets have in common.

***Exercise* #1:** For each of the following, sets A and B are given. Find $A \cup B$ and $A \cap B$.

(a) $A = \{2, 4, 6, 8, 10\}$ $B = \{2, 3, 4, 5, 6\}$

$A \cup B =$

$A \cap B =$

(b) $A=\{\ldots,-3,-2,-1,0,1,2,3,\ldots\}$ $B=\{\ldots-5,-3,-1,1,3,5,\ldots\}$

$A \cup B =$

$A \cap B =$

(c) A = {apples, bananas, oranges, grapefruit, peaches} B = {lemons, peaches, apples}

$A \cup B =$

$A \cap B =$

Skills

1. Complete the following table.

A	B	$A \cup B$	$A \cap B$
$\{H,O,U,S,E\}$	$\{H,O,R,S,E\}$		
$\{1,2,3,4,5,6\}$	$\{1,3,5,7,9\}$		
{Even, Odd}	{Odd}		
{T, A, B, L, E}	{S, I, N, G}		
$\{3,6,9,12,15,18,21,24\}$	$\{4,8,12,16,20,24,28\}$		
$\{x : x > 1\}$	$\{x : x \leq 1\}$		
{1, 3, 6, 10, 15, 21, 28}	{10, 11, 12, 13, 14, 15}		

Applications

2. If $A=\{(1,2),(2,3),(3,4),(4,5),(5,6)\}$ and $B=\{(4,7),(3,5),(2,3),(1,2),(0,-1),(-1,-3)\}$, then find $A \cap B$. (Hint: Your answer should consist of one or more ordered pairs.)

Mixed Review:

1. Which statement represents "all numbers between negative 4 and positive 8"?
[1] $-4 \leq x \leq 8$ [2] $-4 < x < 8$ [3] $-4 \leq x < 8$ [4] $-3 \leq x \leq 7$

2. Which interval notation represents the set of numbers that are greater than or equal to -1, but are less than 9?
[1] (-1,9] [2] [-1,9] [3] (-1,9) [4] [-1,9)

3. Which of the following sets is a subset of {2, 4, 6, 8, 10, 12}?
[1] {14} [2] {2, 3, 4} [3] {4, 8, 12} [4] {1, 3, 5}

4. If the universal set is $U = \{1^2, 2^2, 3^2, 4^2, 5^2, 6^2\}$, what is the complement of the intersection of set $A = \{2^2, 4^2, 6^2\}$ and set $B = \{2^2, 3^2, 4^2\}$?
[1] $\{2^2, 4^2\}$ [2] $\{1^2, 5^2\}$ [3] $\{1^2, 5^2, 6^2\}$ [4] $\{1^2, 3^2, 5^2, 6^2\}$

5. In a sophomore class of 80 students, 42 students take Spanish, 24 take French, and 4 students take both Spanish and French. How many students in this class are not enrolled in either Spanish or French?
[1] 18 [2] 14 [3] 12 [4] 10

6. Given the relation $D = \{(6,4), (8,-1), (x,7), (-3,-6)\}$. Which of the following values for x will make relation D a function?
[1] -3 [2] -6 [3] 8 [4] 6

7. Is the relation depicted in the chart below a function?
[1] Yes [2] No [3] Cannot be determined

x	-2	3	-6	4	-7	5
y	3	-6	3	-6	3	-6

8. If $f(x) = -3x - 5$, what is the value of $f(2)$?
[1] -11 [2] -1 [3] 1 [4] 11

1.________
2.________
3.________
4.________
5.________
6.________
7.________
8.________

9. If $g(x) = 3x^2 - 2x - 5$, what is the value of $g(-1)$?

[1] -4 [2] -10 [3] -6 [4] 0

9.______

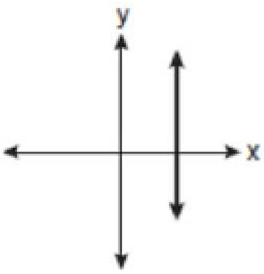
10. Which relation is *not* a function?

[1] {(2,5), (3,6), (4,7), (5,8)} [2] {(6,-2), (-4,6), (-2,4), (1,0)}
[3] {(-1, 5), (-2,5), (-3,5), (-4,5)} [4] {(0,-2), (1,0), (-1,-3), (0,-1)}

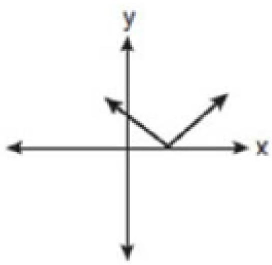
10.______

11. Which graph represents a function?

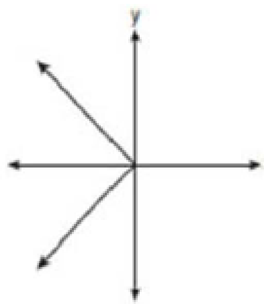
[1]



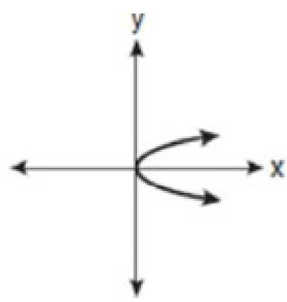
[2]



[3]



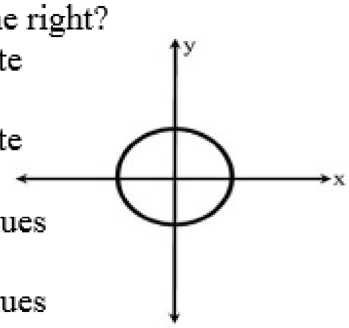
[4]



11.______

12. Which statement is true about the relation shown at the right?

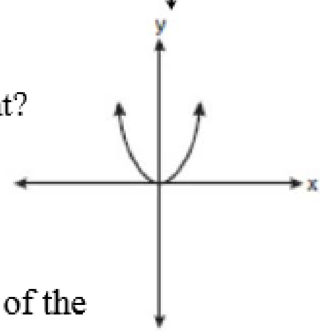
[1] It is a function because there exists one y-coordinate for each x-coordinate.
[2] It is a function because there exists one x-coordinate for each y-coordinate.
[3] It is *not* a function because there are multiple x-values for a given y-value.
[4] It is *not* a function because there are multiple y-values for a given x-value.



12.______

13. What type of function is shown by the graph at the right?

[1] linear [2] exponential
[3] quadratic [4] absolute value



13.______

14. A ball is tossed in the air in such a way that the path of the ball is modeled by the equation $y = -x^2 + 6x$, where y represents the height of the ball in feet and x is the time in seconds. At what time, x, is the ball at its highest point? [1] 6 [2] 2 [3] 3 [4] 4

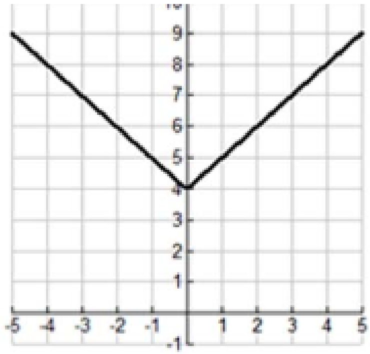
14.______

15. Consider the graph of the function $y = 2^x$. At what point will this graph intersect the x-axis?

[1] (2,0) [2] (0,0) [3] (1,0) [4] Never intersects x-ax

15.______

16. Which equation is represented by this graph?



[1] $y = x^2 + 4$ [2] $y = (x + 4)^2$
[3] $y = |x| + 4$ [4] $y = |x + 4|$

16._______

17. An arch is built in the shape of a parabola given by the equation $y = -\frac{1}{2}x^2 + 4x$. The base of the arch is determined by the two locations where the equation crosses the x-axis. If each unit on the x-axis represents one foot, how wide is the base of the arch?

[1] 4 feet [2] 6 feet [3] 8 feet [4] 10 feet

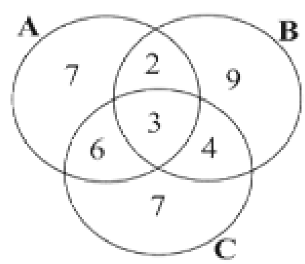
17._______

18. Given sets $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8\}$ and $C = \{2, 3, 4, 5\}$, what is the intersection of the three sets?

[1] {2, 4} [2] {2, 3, 4} [3] {2, 4, 5} [4] {1, 2, 3, 4, 5, 6, 8}

18._______

19. Which set represents the intersection of sets *A, B,* and *C* as shown in the diagram at the right?



[1] {2, 4, 6} [2] {2, 3, 4, 6}
[3] {2, 3, 4, 6, 7, 9} [4] {3}

19._______

20. Which set builder notation describes {-2, -1, 0, 1, 2, 3, 4, 5, 6}?

[1] $\{x \mid -2 \le x \le 6, \text{where } x \text{ is an integer}\}$

[2] $\{x \mid -2 < x < 6, \text{where } x \text{ is an integer}\}$

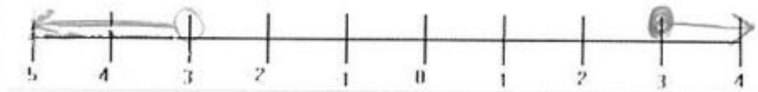
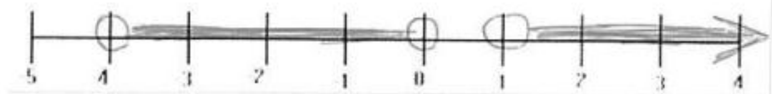
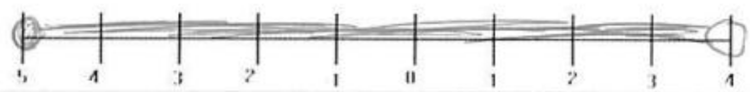
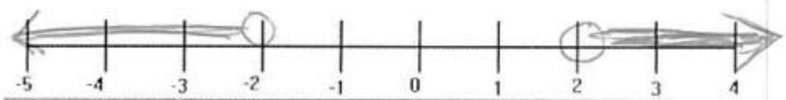
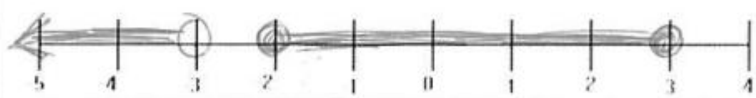
[3] $\{x \mid -2 < x \le 6, \text{where } x \text{ is an integer}\}$

[4] $\{x \mid -2 \le x < 6, \text{where } x \text{ is an integer}\}$

20._______

Answer Key

Practice: Putting it all together:

	Set-builder Notation	Interval Notation	Graph
1.	$\{x \in \mathbb{R} \mid x < -3 \text{ or } x \geq 3\}$	$(-\infty,-3) \cup [3,\infty)$	
2.	$\{x \in \mathbb{R} \mid -4 < x < 0 \text{ or } x > 1\}$	$(-4,0) \cup (1,\infty)$	
3.	$\{x \in \mathbb{R} \mid -4 \leq x \leq -2\}$	$[-4,-2]$	
4.	$\{x \in \mathbb{R} \mid -5 \leq x < 4\}$	$[-5,4)$	
5.	$\{x \in \mathbb{R} \mid x < -2 \text{ or } x > 2\}$	$(-\infty,-2) \cup (2,\infty)$	
6.	$\{x \in \mathbb{R} \mid -1 \leq x < 1 \text{ or } x = 2\}$	$[-1,1) \cup [2]$	
7.	$\{x \in \mathbb{R} \mid x < -3 \text{ or } -2 \leq x \leq 3\}$	$(-\infty,-3) \cup [-2,3]$	
8.	$\{x \in \mathbb{R} \mid x \leq -4 \text{ or } x > -2\}$	$(-\infty,-4] \cup (-2,\infty)$	
9.	$\{x \in \mathbb{R} \mid -1 \leq x \leq 0 \cup 1 < x \leq 2\}$	$[-1,0] \cup (1,2]$	
10.	$\{x \in \mathbb{R} \mid x \leq -2 \text{ or } -1 < x < 4\}$	$[2,\infty) \cup (-1,4)$	

Extra Practice of Complements:

Suppose that set $B = \{1, 2, 3, 4, 5, 6\}$ represents the "given universe" for a certain situation. Suppose, further, that set $A = \{2, 4, 6\}$. Then, we know that A is a subset of B. We express that relationship with symbols as $A \subset B$. The **complement** of set A here is $\{1, 3, 5\}$ because those are the elements in set B that are **not** in set A. In other words, the complement of a subset consists of all elements in the "given universe" that are not in the subset.

***Exercise* #2:** Consider the set E defined as $E = \{2, 3, 5, 7, 11, 13\}$ and set F defined as $F = \{2, 5, 7\}$. Determine the complement of set F within set E. (We denote the complement of set F using many different symbols: $\sim F$, F', or $\overline{F}$.)

$E = \{\cancel{2}, 3, \cancel{5}, \cancel{7}, 11, 13\}$
$F = \{2, 5, 7\}$

$\boxed{\overline{F} = \{3, 11, 13\}}$

***Exercise* #3:** Consider the set of integers as the "given universe." Determine the complement of the set of whole numbers. Express the complement in roster form.

$U = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\}$
Whole numbers: $\{0, 1, 2, 3 \ldots\}$

$\boxed{\text{Complement: } \{\ldots -4, -3, -2, -1\}}$

***Exercise* #4:** What is the complement of the set that represents the "given universe" in any situation?

$\{\ \}$ ← Empty set → no elements are contained in this set

***Exercise* #5:** Given the set of numbers on a standard six sided die, which of the following represents the complement of the set comprised of all multiples of 3?

(1) $\{1, 2, 3\}$ (3) $\{1, 2, 3, 4, 5\}$
(2) $\{2, 4, 6\}$ (4) $\{1, 2, 4, 5\}$

$\{1, 2, \cancel{3}, 4, 5, \cancel{6}\}$
multiples of 3: $\{3, 6\}$ 4
complement: $\{1, 2, 4, 5\}$

Extra Practice with Union and Intersection:

Operations on Sets: Union and Intersection
Algebra 1

We don't "add" two sets together. Nor do we subtract, multiply, or divide them. However, there are two operations on sets that are commonly used. We call the operations **union** and **intersection.** Just as when we use basic operations on numbers we get numbers for answers, when we operate on sets we get sets for answers. Thus, the union of two sets is a set, and the intersection of two sets is a set. The definitions of union and intersection are given below in set-builder notation.

THE UNION AND INTERSECTION OF TWO SETS

For two sets, A and B, their union, $\cup$, and their intersection, $\cap$, are given by:

(1) **Union**: $A \cup B = \{x : x \in A \text{ or } x \in B\}$ (2) **Intersection**: $A \cap B = \{x : x \in A \text{ and } x \in B\}$

The union of two sets can be thought of as the "joining" of the sets (any element in either set must appear in their union). The intersection of two sets consists of whatever elements the two sets have in common.

***Exercise* #1:** For each of the following, sets A and B are given. Find $A \cup B$ and $A \cap B$.

union (a) $A = \{2, 4, 6, 8, 10\}$ $B = \{2, 3, 4, 5, 6\}$

A "OR" B $A \cup B =$ $\{2, 3, 4, 5, 6, 8, 10\}$

Intersection
A "AND" B $A \cap B =$ $\{2, 4, 6\}$

(b) $A=\{\ldots,-3,-2,-1,0,1,2,3,\ldots\}$ $B=\{\ldots-5,-3,-1,1,3,5,\ldots\}$

$A\cup B=\{\ldots -5,-4,-3,-2,-1,0,1,2,3,4,5,\ldots\}$ (all integers)

$A\cap B=\{\ldots -5,-3,-1,1,3,5\ldots\}$ (all odd integers)

(c) A = {apples, bananas, oranges, grapefruit, peaches} B = {lemons, peaches, apples}

all together → $A\cup B$ = {apples, bananas, oranges, grapefruit, peaches, lemons}

what they have in common → $A\cap B$ = {apples, peaches}

Skills

1. Complete the following table.

A	B	$A\cup B$	$A\cap B$
$\{H,O,U,S,E\}$	$\{H,O,R,S,E\}$	$\{H,O,U,R,S,E\}$	$\{H,O,S,E\}$
$\{1,2,3,4,5,6\}$	$\{1,3,5,7,9\}$	$\{1,2,3,4,5,6,7,9\}$	$\{1,3,5\}$
{Even, Odd}	{Odd}	{even, odd}	{odd}
{T, A, B, L, E}	{S, I, N, G}	{T, A, B, L, E, S, I, N, G}	{ }
$\{3,6,9,12,15,18,21,24\}$	$\{4,8,12,16,20,24,28\}$	$\{3,4,6,8,9,12,15,16,18, 20,21,24,28\}$	$\{12,24\}$
$\{x: x>1\}$ $\{2,3,4,5\ldots\}$	$\{x: x\le 1\}$ $\{\ldots -3,-2,-1,0,1\}$	$\{\ldots -3,-2,-1,0,1,2\ldots\}$	{ }
{1, 3, 6, 10, 15, 21, 28}	{10, 11, 12, 13, 14, 15}	{1, 3, 6, 10, 11, 12, 13, 14, 15, 21, 28}	{10, 15}

Applications

2. If $A=\{(1,2),(2,3),(3,4),(4,5),(5,6)\}$ and $B=\{(4,7),(3,5),(2,3),(1,2),(0,-1),(-1,-3)\}$, then find $A\cap B$. (Hint: Your answer should consist of one or more ordered pairs.)

$\boxed{A\cap B=\{(1,2),(2,3)\}}$

Mixed Review:

1. Which statement represents "all numbers between negative 4 and positive 8"?
   [1] $-4 \le x \le 8$ [2] $-4 < x < 8$ [3] $-4 \le x < 8$ [4] $-3 \le x \le 7$

1. 2

2. Which interval notation represents the set of numbers that are greater than or equal to -1, but are less than 9? not included
   included
   [1] (-1,9] [2] [-1,9] [3] (-1,9) [4] [-1,9)

2. 4

3. Which of the following sets is a subset of {2, 4, 6, 8, 10, 12}?
   [1] {14} [2] {2, 3, 4} [3] {4, 8, 12} [4] {1, 3, 5}

3. 3

4. If the universal set is $U = \{1^2, 2^2, 3^2, 4^2, 5^2, 6^2\}$, what is the complement of the intersection of set $A = \{2^2, 4^2, 6^2\}$ and set $B = \{2^2, 3^2, 4^2\}$?
   [1] $\{2^2, 4^2\}$ [2] $\{1^2, 5^2\}$ [3] $\{1^2, 5^2, 6^2\}$ [4] $\{1^2, 3^2, 5^2, 6^2\}$

   $A \cap B = \{2^2, 4^2\}$ $\overline{A \cap B} = \{1^2, 3^2, 5^2, 6^2\}$

4. 4

5. In a sophomore class of 80 students, 42 students take Spanish, 24 take French, and 4 students take both Spanish and French. How many students in this class are not enrolled in either Spanish or French?
   [1] 18 [2] 14 [3] 12 [4] 10

   Spanish French
   38 (4) 20

   $80 - 62 = 18$

   38
   20
   +4
   62

5. 1

6. Given the relation $D = \{(6,4), (8,-1), (x,7), (-3,-6)\}$. Which of the following values for x will make relation D a function? cannot be 6, 8 or -3
   [1] -3 [2] -6 [3] 8 [4] 6

6. 2

7. Is the relation depicted in the chart below a function?
   [1] Yes [2] No [3] Cannot be determined

each input has exactly one output

x	-2	3	-6	4	-7	5
y	3	-6	3	-6	3	-6

7. 1

8. If $f(x) = -3x - 5$, what is the value of $f(2)$?
   [1] -11 [2] -1 [3] 1 [4] 11

$f(2) = -3(2) - 5$
$f(2) = -6 - 5$
$f(2) = -11$

8. 1

9. If $g(x) = 3x^2 - 2x - 5$, what is the value of $g(-1)$?

[1] -4 [2] -10 [3] -6 [4] 0

$g(-1) = 3(-1)^2 - 2(-1) - 5$
$g(-1) = 3(1) + 2 - 5$
$g(-1) = 3 + 2 - 5$
$g(-1) = 5 - 5$
$g(-1) = 0$

9. 4

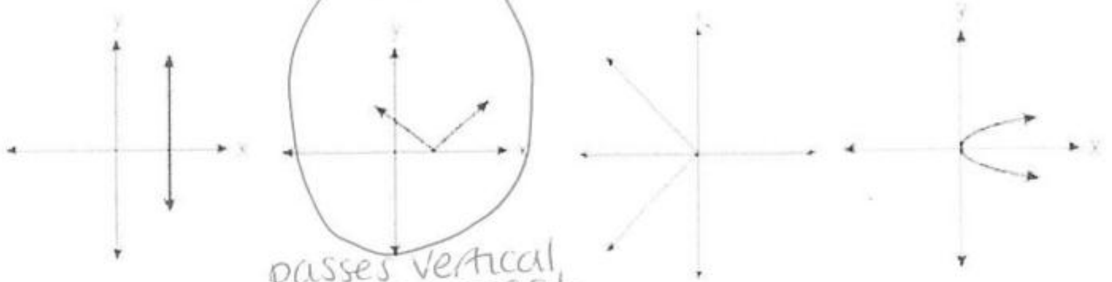
10. Which relation is *not* a function?

[1] {(2,5), (3,6), (4,7), (5,8)} [2] {(6,-2), (-4,6), (-2,4), (1,0)}
[3] {(-1, 5), (-2,5), (-3,5), (-4,5)} [4] {(0,-2), (1,0), (-1,-3), (0,-1)}

each input does NOT have exactly one output.

10. 4

11. Which graph represents a function?

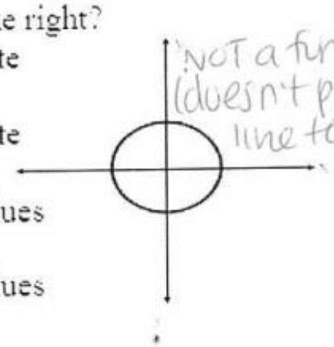


passes vertical Line Test

11. 2

12. Which statement is true about the relation shown at the right?

~~[1]~~ It is a function because there exists one y-coordinate for each x-coordinate.
~~[2]~~ It is a function because there exists one x-coordinate for each y-coordinate.
[3] It is *not* a function because there are multiple x-values for a given y-value. same y
[4] It is *not* a function because there are multiple y-values for a given x-value. (2,4)



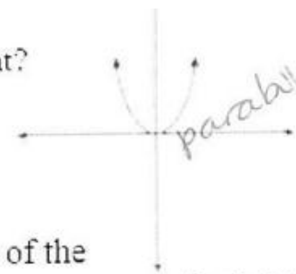
NOT a function (doesn't pass vertical line test)

same x (2,-4)

12. 4

13. What type of function is shown by the graph at the right?

[1] linear [2] exponential
[3] quadratic [4] absolute value



parabola

13. 3

14. A ball is tossed in the air in such a way that the path of the ball is modeled by the equation $y = -x^2 - 6x$, where y represents the height of the ball in feet and x is the time in seconds. At what time, x, is the ball at its highest point? [1] 6 [2] 2 [3] 3 [4] 4

X	Y
0	0
1	5
2	8
3	9
4	8
5	5
6	0

14. 3

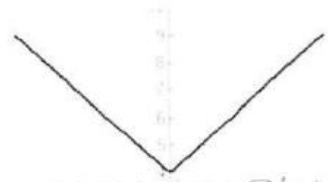
15. Consider the graph of the function $y = 2^x$. At what point will this graph intersect the x-axis?

[1] (2,0) [2] (0,0) [3] (1,0) [4] Never intersects x-ax

approaches x-axis, will not intersect it.

15. 4

16. Which equation is represented by this graph?



~~[1]~~ $y = x^2 - 4$ ~~[2]~~ $y = (x + 4)^2$
[3] $y = |x| - 4$ [4] $y = |x - 4|$

4 units up 4 units left

absolute value
Shifted UP 4 units
from: $y = |x|$

16. 3

17. An arch is built in the shape of a parabola given by the equation $y = -\frac{1}{2}x^2 - 4x$. The base of the arch is determined by the two locations where the equation crosses the x-axis. If each unit on the x-axis represents one foot, how wide is the base of the arch?

x	y
0	0
1	3.5
2	6
3	7.5
4	8
5	7.5
6	6
7	3.5
8	0

[1] 4 feet [2] 6 feet [3] 8 feet [4] 10 feet

17. 3

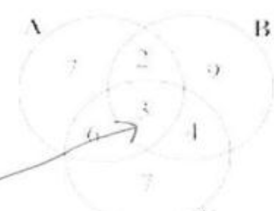
18. Given sets $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8\}$ and $C = \{2, 3, 4, 5\}$, what is the intersection of the three sets?

[1] {2, 4} [2] {2, 3, 4} [3] {2, 4, 5} [4] {1, 2, 3, 4, 5, 6, 8}

18. 1

19. Which set represents the intersection of sets A, B, and C as shown in the diagram at the right?

[1] {2, 4, 6} [2] {2, 3, 4, 6}
[3] {2, 3, 4, 6, 7, 9} [4] {3}



$A \cap B \cap C$

19. 4

20. Which set builder notation describes {-2, -1, 0, 1, 2, 3, 4, 5, 6}?

[1] $\{x \mid -2 \le x \le 6$, where x is an integer$\}$
[2] $\{x \mid -2 < x < 6$, where x is an integer$\}$
[3] $\{x \mid -2 < x \le 6$, where x is an integer$\}$
[4] $\{x \mid -2 \le x < 6$, where x is an integer$\}$

-2 and 6 ARE INCLUDED.

20. 1